

EE3621: Digital Signal Processing

Lecture 18: IIR Cascade Realization & Second-Order Sections (SOS) (III B.Tech EEE)

September 9, 2026

1. LEARNING OBJECTIVES

By the end of this lecture, students will be able to:

1. **Factor** high-order IIR transfer functions into second-order biquad sections (SOS).
 2. **Apply** pole-zero pairing rules to minimize frequency response peaking.
 3. **Order** biquad sections to optimize signal-to-noise ratio (SNR) and prevent overflow.
 4. **Draw** complete Cascade realization signal flow graphs.
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2. MATHEMATICAL FOUNDATIONS

2.1 The Cascade Realization Formulation

$$H(z) = g \prod_{k=1}^K H_k(z) = g \prod_{k=1}^K \frac{b_{k0} + b_{k1}z^{-1} + b_{k2}z^{-2}}{1 + a_{k1}z^{-1} + a_{k2}z^{-2}}, \quad K = \lceil N/2 \rceil$$

Each biquad $H_k(z)$ realizes one real pole pair and one real zero pair:

$$1 + a_{k1}z^{-1} + a_{k2}z^{-2} = (1 - p_k z^{-1})(1 - p_k^* z^{-1}) = 1 - 2r_k \cos(\theta_k)z^{-1} + r_k^2 z^{-2}$$

2.2 Pairing and Ordering Rules

1. Pole-Zero Pairing:

* Pair the complex pole pair p_k having highest Q (closest to unit circle) with the zero pair z_k closest to it in angle θ . * This cancels large resonance peaks, keeping the section gain $|H_k(e^{j\omega})|$ relatively flat.

1. Section Ordering:

* Arrange sections in order of **increasing pole radius** r_k (increasing Q / peaking). * Low- Q sections first $\rightarrow$ High- Q sections last. * Prevents premature signal saturation in early stages.

Pictorial Representations

Cascade (Serial) IIR realization (SOS 1 followed by SOS 2)

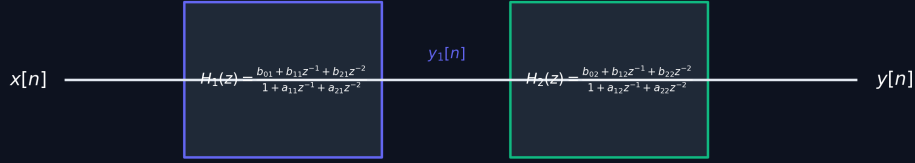


Figure 1: Iir Cascade

Parallel IIR realization (SOS channels added in parallel)

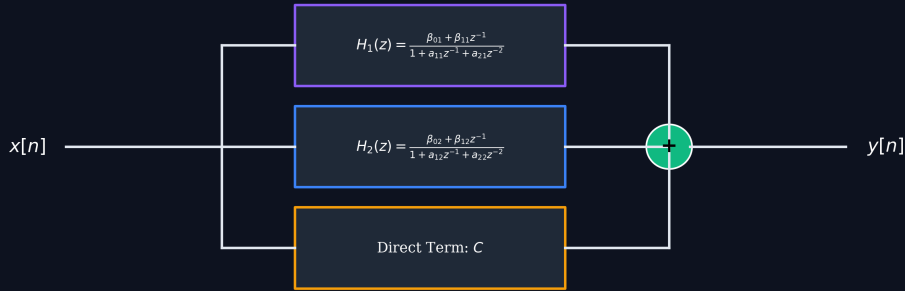


Figure 2: Iir Parallel

3. WORKED NUMERICAL EXAMPLES

Example 18.1: 4th-Order IIR Cascade Realization

Problem: Realize $H(z) = \frac{(1+z^{-1})(1-\sqrt{2}z^{-1}+z^{-2})}{(1-0.9e^{j\pi/4}z^{-1})(1-0.9e^{-j\pi/4}z^{-1})(1-0.6z^{-1})(1+0.5z^{-1})}$ in Cascade Form with optimal pairing.

Solution:

1. Poles:

* Complex pair: $p_{1,2} = 0.9e^{\pm j\pi/4} \implies 1 - 2(0.9)\cos(\pi/4)z^{-1} + 0.81z^{-2} = 1 - 1.2728z^{-1} + 0.81z^{-2}$. (Radius $r = 0.9$, high Q).
 * Real poles: $p_3 = 0.6$, $p_4 = -0.5 \implies (1 - 0.6z^{-1})(1 + 0.5z^{-1}) = 1 - 0.1z^{-1} - 0.3z^{-2}$. (Low Q).

1. Zeros:

* Complex pair: $z_{1,2} = 1 \cdot e^{\pm j\pi/4} \implies 1 - \sqrt{2}z^{-1} + z^{-2} = 1 - 1.4142z^{-1} + z^{-2}$ (at angle $\pi/4$).
 * Real zeros: $z_3 = -1 \implies (1 + z^{-1})$.

1. Optimal Pairing:

* Pair poles at angle $\pi/4$ ($p_{1,2}$) with zeros at angle $\pi/4$ ($z_{1,2}$). * Section 1 (Low Q): $H_1(z) = \frac{1+z^{-1}}{1-0.1z^{-1}-0.3z^{-2}}$. * Section 2 (High Q): $H_2(z) = \frac{1-1.4142z^{-1}+z^{-2}}{1-1.2728z^{-1}+0.81z^{-2}}$.

1. **Ordering:** Section 1 followed by Section 2.

4. UNIVERSITY EXAMINATION QUESTIONS & MARKING RUBRIC

Question 1 (15 Marks)

(a) Why are direct form structures not suitable for implementing high-order IIR filters? Explain coefficient quantization sensitivity. *(6 Marks)* (b) Realize the transfer function in Cascade Form:

$$H(z) = \frac{1 + \frac{1}{3}z^{-1}}{(1 - \frac{1}{2}z^{-1})(1 - \frac{1}{4}z^{-1} + \frac{1}{2}z^{-2})}$$

Draw the complete signal flow graph using Direct Form II biquads. *(9 Marks)*

Model Answer & Step-by-Step Marking Rubric:

- **Part (a):**

* Root sensitivity explanation: Roots of $A(z)$ depend non-linearly on polynomial coefficients. In high-order polynomials, 1% coefficient perturbation can displace poles by 50% or move them outside $|z| = 1$ *(6 Marks)*

- **Part (b):**

* Split into 1st-order and 2nd-order sections: $H_1(z) = \frac{1+\frac{1}{3}z^{-1}}{1-\frac{1}{2}z^{-1}}$, $H_2(z) = \frac{1}{1-\frac{1}{4}z^{-1}+\frac{1}{2}z^{-2}}$ *(4 Marks)* * Neatly drawn Cascade SFG of $H_1(z)$ cascaded into $H_2(z)$ using canonical DF-II sections *(5 Marks)*

5. PYTHON VERIFICATION SCRIPT

```
import scipy.signal as signal
import numpy as np

b = [1, 1/3]
a = np.convolve([1, -0.5], [1, -0.25, 0.5])
sos = signal.tf2sos(b, a)
print("Computed SOS:")
print(sos)
```